AI2002 – Artificial Intelligence
Quiz – 5A

Name	
Roll. No.	
Section	

CLO2: Apply classical AI techniques, including uninformed and informed search algorithms, minimax, and neural networks, to solve structured problems.

Q1. [6 Marks] Given the following training points for two features (x1, x2) and their classes, classify the new test point T = (4, 3). Show all working. Round L2 distances to 2 decimal places.

Point	x1	x2	Class
P1	2	4	R
P2	5	8	B
P3	8	2	B
P4	1	1	R

a) Using L1 (Manhattan) distance, what is the predicted class for k = 1?
b) Using L2 (Euclidean) distance, what is the predicted class for k = 3?

Q2. [4 Marks] A dataset contains 15,000 images. It is split into 60% training, 20% validation, and 20% testing.
a) How many images are in the validation set?
b) Assume a standard 3-NN implementation computes the distance from each test image to every training image before selecting the 3 nearest. If one distance computation takes 0.05 ms, how many minutes will it take to classify the entire test set?

Solutions

Q1 Solution

L1 (Manhattan) distances from T:

Point	Coordinates	Class	Distance
P1	(2, 4)	R	3.00
P2	(5, 8)	B	6.00
P3	(8, 2)	B	5.00
P4	(1, 1)	R	5.00

Nearest point for part (a): P1 (R). Predicted class = R.

L2 (Euclidean) distances from T:

Point	Coordinates	Class	Distance
P1	(2, 4)	R	2.24
P2	(5, 8)	B	5.10
P3	(8, 2)	B	4.12
P4	(1, 1)	R	3.61

Three nearest points for part (b): P1 (R), P4 (R), P3 (B). Predicted class = R.

Q2 Solution

a) Validation images = 20% of 15,000 = 3,000.

b) Training images = 9,000; Test images = 3,000.

Total distance computations = 3,000 × 9,000 = 27,000,000.

Total time = 27,000,000 × 0.05 ms = 1,350,000 ms = 22.50 minutes.

Checked final answers:

Q1(a): R | Q1(b): R | Q2(a): 3,000 | Q2(b): 22.50 minutes